TRUNKBELAY · DESIGN DECISION RECORD

TrunkBelay TRL 2 review decisions

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-03	Amish Chadha	TRL 2 review items decided under Amish's 2026-10-03 pre-approvals

0001: TRL 2 review decisions

- **Date:** 2026-10-03
- **Status:** decided, except where noted. Decided by Amish under his pre-approval of 2026-10-03: "start with the first 14 repos from the list of 29 projects. I pre-approve the batch runs along with any recommendations you come up with. I also accept any cost overruns or variations from the assumed scope cost." and his instruction of the same day for this batch: "Proceed with the remaining 15 scaffolds". Requirements that are not met or at risk are not decided here; they are open in the design decisions register (TKB-DEC-001), as Amish asked on 2026-10-03: "A simple statement doesn't add value - ensure you are identifying a state and posing it as a clear recommendation for me to decide on."

Context

The scaffold (TKB-PRB-001, TKB-PRC-001 and TKB-REQ-001, all v0.1) described a chain collar with a slotted cleat, a padded bearing plate, a commodity friction device, a rope, a victim sling and a bag. It left five open questions: the load turning from upward to downward as the victim passes the collar, grip on wet and curved trunks, which friction device to use, how to fit a sling to an unconscious person hanging head down, and whether fire services would adopt the kit. Populating the concept to TRL 2 meant settling these on paper. Items that touch safety take the conservative option and say what evidence would relax it. Partners are the first candidates to approach, not agreements.

Options considered

Table 1. Options.

#	Item	Options
D1	How the collar grips	(a) a band tightened hard round the trunk (grip set by the rescuer's pull); (b) a self-locking frame, the load hung on a lever so the grip rises with the load
D2	Bearing on the bark	(a) a curved padded plate; (b) two rubber-faced bars in a V
D3	Where the collar goes	(a) level with the rescuer, below the victim, and loaded upward at first; (b) at least 300 mm above the point where the rope meets the victim, always loaded downward
D4	Friction device	(a) a figure-eight descender; (b) a bar rack; (c) a Munter hitch on a carabiner; (d) an auto-locking descender with anti-panic, plus a brake carabiner
D5	Victim attachment	(a) one sewn sling as a hasty seat; (b) an evacuation triangle with a chest sling
D6	Rope and spare rope	(a) all rope in a bag hung on the collar; (b) the bag lowered to the helpers and the brake strand run to the ground
D7	Rating	(a) a two-person load for fire crews; (b) one person, 100 kg design mass, sized for a 2.5 kN static load (R1)
D8	First co-design candidates	Climber training programme, fire station, climbers' collective
D9	Material	(a) steel, welded and hot-dip galvanised; (b) aluminium

Decision

Table 2. Decisions.

#	Decision	Why
D1	(b) Self-locking frame. The load eye sits 150 mm out from the frame and the chain line 160 mm above its foot, so the frame cocks on the trunk	Grip does not depend on how hard a tired rescuer pulls; it rises with the load (TKB-CAL-001, B1 to B4)
D2	(b) Two rubber-faced bearing bars in a 120 deg V	Touches every trunk from 200 to 450 mm on two lines and centres the frame; a curved plate fits one size
D3	(b) for the drill as modelled. Whether this answers R2 or the collar is made to hold both ways is open: TKB-DEC-001, O1	The single top chain cannot hold an upward pull on paper
D4	(d) Auto-locking descender with anti-panic, brake strand over a carabiner on the spare eye	Conservative: it holds the victim if the rescuer lets go or pulls too hard. Relaxed only by a training trial showing that rescuers control a manual device reliably
D5	(b) Evacuation triangle round the hips and a chest sling on the same victim carabiner	Turns the victim head up once the feet are freed and holds an unconscious person
D6	(b) Bag lowered, brake strand to the ground	The rescuer handles one strand at height; the helpers manage the spare rope
D7	(b) One person, 100 kg, 2.5 kN static. No use on a person before a proof test at 2.5 kN on cut trunk sections and a dummy drill at low height	Conservative gate; the rating is raised only by a test programme with a fire service partner
D8	First candidates to approach (none approached yet): a coconut climber training programme under the Coconut Development Board's Friends of Coconut Tree scheme in Kerala; a Kerala Fire and Rescue Services station in Kasaragod or Kannur; a climbers' collective or labour bank in the same district	They teach, rescue and climb
D9	(a) Steel, welded and galvanised, for the prototype. An aluminium frame is one of the options for R7 in TKB-DEC-001, O3	Any village fabrication shop can make it
D10	Pitch, problem, patent design-arounds and `budget_usd` unchanged	Scope rule

Consequences

- The cleat is a pair of plain slots with a keeper and pin, not a ratchet, which keeps the design-around of the 3M Cynch-Lok.
- The kit carries more than its 5 kg target and costs more than its USD 100 target; both are open decisions for Amish (TKB-DEC-001, O3 and O4).
- The drill's placement rule (D3) and rigging time are open decisions for Amish (TKB-DEC-001, O1 and O2).

> **Safety:** D3, D4 and D7 set safety limits. Each takes the conservative option and names the evidence that would relax it. Nobody is lowered with this kit until the collar has been proof-tested at 2.5 kN on cut trunk sections and the drill has been run with a dummy at low height.